DOCUMENT 06

Manufacturing Strategy & Commercialization Roadmap

Full BOM · PCB Architecture · Assembly · Cost-Down · Market Strategy · Business Model

Complete production and commercialization blueprint covering full production BOMs for all three drone classes, custom PCB architecture for fleet-specific electronics, step-by-step assembly procedures, cost-down roadmap from prototype to volume, detailed market segment analysis, go-to-market strategy, revenue model projections, and regulatory compliance pathway.

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1. PRODUCTION BOM — WORKER W1 (STRIPPED)

The complete production BOM for the endurance-optimised W1 worker drone. All mass values are measured; all costs are current as of May 2026.

#	Component	Part / Spec	Qty	g	Unit $	Total $
01	CFRP Frame	Tarot TL65B44, 650mm, [0/±45/90]s	1	320	85	85
02	Motor	T-MOTOR F90-1300KV outrunner	4	144	38	152
03	FOC ESC	T-MOTOR F55A Pro II, BLHeli_32	4	160	72	288
04	Propellers CW+CCW	HQProp 9×4.5×3 carbon + 2 spare sets	8	36	4	32
05	Flight Controller	Holybro Pixhawk 6C, STM32H743	1	60	110	110
06	GPS/Compass	Holybro M9N u-blox M9N + HMC5983	1	36	45	45
07	UWB Module	Decawave DWM3000 EVB	1	10	35	35
08	Li-ion Battery	4S 10,000 mAh 18650 ~148 Wh, 1.10 kg	1	1100	145	145
09	Power Module	Holybro PM02D isolated 5V/4A	1	12	29	29
10	LoRa Telemetry	Holybro SiK 915 MHz 100mW	1	15	35	35
11	ELRS Receiver	ExpressLRS EP2 2.4GHz <1ms	1	5	18	18
12	IR Dock RX PCB	Custom: 4×TEMD6010FX01 + STM32G031	1	12	28	28
13	Docking Keel Rail	CNC Al 6061-T6 anodised black, ~80mm	1	55	45	45
14	NdFeB Magnets	N52 10×3mm 4kg pull ×4	4	8	2	8
15	Pogo Pins (power)	Gold-plated 10A spring-loaded ×4	4	6	3	12
16	Custom PDB	ESC dist + isolated 5V + current sense	1	28	45	45
17	ESC Heatsinks	Cu 40×20×10mm + 5W/mK TIM	4	32	3	12
18	Vibration Mounts	M3 rubber grommet isolator ×8	8	8	1	5
19	Wiring Harness	12AWG silicone, XT60/XT90, JST-GH	1 set	35	18	18
20	Thermal Fan (hot climate)	40mm 5V 0.3W axial	1	12	5	5
21	Fasteners + Hardware	M2/M3 titanium alloy + threadlock	1 set	12	12	12
22	Conformal Coat	Acrylic spray 400ml (shared across 5 drones)	0.2	4	3	3
	PARTS SUBTOTAL			**~2,110 g**		**$1,170**

#	Component	Part / Spec	Qty	g	Unit $	Total $
	Assembly labour 2.5hr @ $75/hr					$187
	Calibration & test					$50
	Spare parts (10% contingency)					$117
	TOTAL PER UNIT (single)					**$1,524**
	TOTAL PER UNIT (100-unit batch)					**$890**

Table 1. Worker W1 production BOM. Target MTOW 1,650 g at optimal battery mass fraction.

2. PRODUCTION BOM — COMMAND NODE C1

Component	Specification	g	Unit $
CFRP Frame (heavy-lift variant)	Tarot T810 or custom 700mm frame	420	120
Motor ×6	T-MOTOR MN4014 400KV heavy lift	6×150=900	6×75=450
ESC ×6	T-MOTOR Alpha 60A FOC	6×65=390	6×90=540
Propeller ×6	T-MOTOR 15×5 carbon folding	6×90=540	6×35=210
Flight Controller	Holybro Pixhawk 6C (same as worker)	60	110
Jetson Orin NX 16GB	100 TOPS, 25W TDP, JetPack 6.0	158	499
Jetson Baseboard	Holybro Pixhawk-Jetson, isolated PSU, CAN	45	95
Heatsink + Fan	40×40 8-fin passive + 40mm active 5V 0.3W	40	23
256GB microSD	Kingston Industrial UHS-II	2	45
Intel RealSense D435i	RGB-D 30fps, factory calibrated intrinsics	72	179
Livox Mid-360 LiDAR	360°×59° 200k pts/s, Ethernet UDP	265	599
RTK GPS F9P	Holybro H-RTK F9P Helical dual-antenna	100	240
RTK Boom	50cm CFRP boom, 2× patch antennas	45	60
LoRa 500mW	Holybro SiK 900MHz 500mW	20	65
High-gain antenna	900MHz dipole 5dBi	30	28
ESP32 mesh hub	ESP32-S3 module, ESP-NOW hub firmware	8	10
LiPo battery	6S 6,000mAh 45C (450g, 89Wh)	450	70
Power Module	Holybro PM02D isolated (×2 total for C1)	24	58
Wiring + misc	Full harness, connectors, IP54 sealing	80	45
PARTS SUBTOTAL		**~3,669g**	**~$3,446**
Assembly labour 4hr @ $75/hr			$300
Calibration (LiDAR align, RTK setup)			$150

Component	Specification	g	Unit $
TOTAL C1 (single)			**$3,896**
TOTAL C1 (10-unit batch)			**$2,800**

Table 2. Command node C1 BOM and cost.

3. CUSTOM PCB ARCHITECTURE

3.1 Custom Power Distribution Board (PDB-01)

The PDB-01 consolidates four separate current-production components into one 60×60mm 4-layer board: (1) main ESC power distribution, (2) per-motor current sensing for telemetry, (3) isolated 5V/4A FC rail replacing PM02D, and (4) isolated 5V/3A compute rail. Mass saving: 85g → 28g (–57g). Connector saving: eliminates 6 intermediate XT60/JST connectors that are common failure points. Cost at JLCPCB 50-unit run: $28/board vs $87 for separate components.

Block	Function	Key ICs	Test Point
XT90 main input + fusing	100A blade fuse + main bus	Littelfuse 100A	Volt meter pad
4× XT60 ESC outputs	ESC distribution bus bars	Heavy trace 35A each	Current per output
Per-motor current sense	4× shunt + amplifier for MAVLink	4× INA226 I^{2}C	I^{2}C bus to FC
Isolated 5V/4A FC buck	Decoupled FC + servo power	TI TPS82130 or TPSM82866	Oscilloscope isolation check
Isolated 5V/3A compute	Decoupled Jetson/companion power	TI TPSM82866	Load regulation ±1%
Voltage ADC	Full-pack and cell-tap voltage	ADS1115 16-bit I^{2}C	Calibration against reference
Temperature sense ×4	ESC thermal zones	NTC10K on ESC pad areas	Thermal camera correlation

Table 3. PDB-01 functional block diagram.

3.2 IR Docking Guidance PCB (IRDG-01)

The IRDG-01 is a 40×40mm 4-layer board mounted flush on the drone underbelly. It contains the complete drone-side IR guidance electronics: four TEMD6010FX01 silicon photodiodes (940nm peak, matched sensitivity, 11ns rise time), four OPA380 transimpedance amplifiers (1MHz bandwidth, selectable gain), four synchronous demodulator circuits (lock-in amplifier topology, one per LED frequency), and an STM32G031K8 MCU producing the computed [ε_x, ε_y, height] tuple at 500Hz over UART to the flight controller.

Component	Function	Key Spec	Notes
TEMD6010FX01 ×4	IR photodetection	940nm, 11ns, SMD package	Matched ±5% sensitivity; 90° half-angle
OPA380AIDBVT ×4	Transimpedance amplifier	1MHz GBP, CMOS input	Gain resistor selectable 10k–1MΩ via 0402 pad
Synchronous demodulators ×4	Lock-in at f■–f■	11/17/23/31 kHz	STM32G0 timer output drives reference signal
STM32G031K8T6	Processing + UART output	48MHz, 8KB RAM, FPU	Custom firmware: 500Hz UART, 4ch demod, error calc

Component	Function	Key Spec	Notes
LP2985 LDO	3.3V power	150mA, tiny SOT-23	Clean supply for analog front end
JST-GH 6-pin	FC connection	GND, 3.3V, UART TX/RX, sync, trigger	Mounts to Pixhawk UART port via adapter

Table 4. IRDG-01 component list.

3.3 Dock Beacon PCB (DOCB-01)

The DOCB-01 dock-side beacon board (60×60mm, IP65-encapsulated) drives the four IR LED emitters at their designated frequencies. Each LED is driven by a half-bridge MOSFET pair generating a precision square wave at the assigned frequency. An STM32G031 generates all four timing signals from a single 48MHz system clock to ensure phase coherence and prevent beat-frequency artefacts.

Component	Function	Key Spec
OSRAM SFH4557 ×4	IR LED emitters, 940nm	100mW radiant power, 20° half-angle, SMD
IRLML6401 MOSFET ×8 (2 per LED)	Half-bridge LED drive	20V, 3.7A, SOT-23
STM32G031K8T6	Timing generation, self-test	48MHz; generates f■=11kHz, f■=17kHz, f■=23kHz, f■=31kHz
PC817 optocoupler	Dock-to-drone ground isolation	If needed for long cables
100µF 10V ceramic ×4	LED bypass decoupling	Prevent supply droop at switching

Table 5. DOCB-01 dock beacon PCB component list.

4. ASSEMBLY PROCEDURE — WORKER DRONE W1

Assembly is divided into five stages. Total time: 2.5 hours per unit at single-technician pace.

Stage 1 — Frame Assembly (25 min)	Mount arm tubes to central frame plate with M3 Ti screws; apply Loctite 243 (blue) to all threads. Install M3 rubber vibration isolator grommets at four FC mounting points (critical: correct orientation). Route main power cable (12 AWG) through each arm channel. Secure cables with nylon P-clips at 80mm intervals. Apply conformal coat to all exposed PCB pads on the frame before assembly. **QC check:** arm alignment ±0.3° with digital angle gauge; all cable routes clean.
Stage 2 — Propulsion (35 min)	Mount T-MOTOR F90 motors to arm tips with M3 screws (4× per motor); verify CW/CCW for X-quad geometry before mounting. Apply thermal interface material (5 W/m·K) to ESC mounting pad on heatsink. Solder ESC three-phase wires to motor (match colour coding to thrust direction). Tin XT60 connectors on ESC power leads. Mount ESC+heatsink assembly to arm (double-sided thermal tape + M2 screws). **QC check:** manual motor rotation — smooth, no grinding; ESC current draw <0.5A at idle.

Stage 3 — Electronics Bay (55 min)	Mount custom PDB to central frame; connect battery XT90 and ESC XT60 wires. Mount Pixhawk 6C on vibration isolators; route FC power cable to PDB isolated rail. Connect GPS to FC UART2; compass to I²C; mount GPS on top-plate mast. Connect UWB DWM3000 to FC SPI2; mount at arm-body junction for clear 360° view. Mount IRDG-01 IR board to belly plate; align photodetectors to keel longitudinal axis ±2°. Mount docking keel rail to belly with M3 bolts; torque 1.5 N·m; install magnets in keel slots. Secure battery with 3M dual-lock + retention strap; verify CG ±5mm of motor plane centroid. **QC check:** power-on BIT: all FC sensor reads nominal; ESC communications verified.
Stage 4 — Calibration (30 min)	Accelerometer calibration (6-position) in QGroundControl. Compass calibration (rotation sequence); verify consistent heading ±5° vs reference. ESC calibration via BLHeli Suite Configurator: verify DSHOT600 protocol, motor rotation, FOC mode enabled, current sensor calibrated. IRDG-01 self-test: send UART query 0x01; verify response with 4-channel intensity values. Battery SOC calibration: charge to 100%, read cell voltages; verify EKF SOC = 100% ±2%. **QC check:** All calibration logs saved to drone build file.
Stage 5 — Functional Test (25 min)	Motor spin test without props: all four CW/CCW correct per quadrant; no abnormal current draw. Vibration bench test with props (test stand, safety cage): run to hover RPM; verify IMU RMS ≤0.5 mm/s. RTH trigger test: set battery failsafe to 80%, verify RTH activates correctly. IR beacon bench test: hover drone over bench dock at 200mm; verify [ε_x, ε_y] = [0,0] ±2mm. Data link test: verify LoRa + ELRS links active at 100m range. **QC check:** Sign-off sheet with technician serial, all checks passed, build notes.

5. COST-DOWN ROADMAP

Stage	Key Actions	Unit Cost Target	Timeline	Savings Driver
Prototype (1–5 units)	Off-shelf components; manual assembly; test-learn	~$1,524/unit	Month 0–3	Learning and de-risking
Pilot batch (10 units)	Bulk T-MOTOR pricing (–10%); custom PDB-01 v1.0; IR PCB from JLCPCB	~$1,050/unit	Month 4–8	Custom PCBs +bulk purchasing
Small production (50 units)	Direct factory pricing T-MOTOR/Holybro (–20%); CNC keel in batch; Li-ion cell direct import	~$820/unit	Month 9–18	Factory sourcing + tooling
Volume (100 units)	Assembly line (–60% labour); PCB integration MCU board replaces Pixhawk+modules; own CFRP mould	~$650/unit	Month 18–30	Assembly line + integration
Scale (500 units)	Custom motor windings; OEM ESC; injection-moulded keel; battery cell direct from CATL	~$450/unit	Month 30–48	Full vertical integration

Table 6. Cost-down roadmap. Target $450/unit at 500-unit scale vs $1,524 prototype.

6. MARKET SEGMENTATION — DEEP ANALYSIS

Segment	Key Pain Point	Our Solution	Competition	Entry Point	TAM (est.)
Power line / pipeline inspection	Multi-hour BVLOS; no persistent single drone	K2 + 5 workers; 7hr coverage; FOC low noise near populated areas	DJI Matrice 350 (38min); Skydio 2 (23min)	Utility company drone programme director	~$800M India; ~$3B global
Precision agriculture	Wide area (>100 ha); multiple flights; multispectral	Modular multispectral payload on W1 rail; fleet coverage	DJI Agras T40 (heavy VTOL); Parrot Sequoia	Ag-drone pilot / agronomist	~$1.2B India; ~$6B global
Search and rescue	Multi-hour persistence; thermal + visual; rapid deploy	K1 portable + W1 workers + thermal module swap	Flyability Elios 3 (30min indoor); DJI Mavic 3T	State/district disaster management	~$200M India; ~$500M global
Telecom infrastructure	Persistent aerial relay node at altitude	Fixed-wing tethered worker as relay; fleet provides coverage	Commercial tethered drones; satellite	Jio/Airtel network engineering	~$150M India (5G rural coverage)
Border/coastal monitoring	24/7 persistent; all-weather; long-range	K2 relay + 5 workers + night vision module	DRDO specific programs; Zipline fixed-wing	Ministry of Defence via tender	Defence — restricted; high margin
Smart city / logistics	Low noise; urban ops; lightweight	W1 at 41dB (urban ambient comparable); modular payload	Wing/Amazon PrimeAir (fixed-wing); Swiggy drones (single purpose)	Urban local body; logistics cos.	Emerging; regulatory risk high

Table 7. Market segment deep analysis. TAM = Total Addressable Market.

7. GO-TO-MARKET STRATEGY

7.1 Phase 1 — Prove Value with Paying Pilots (Month 0–12)

Identify 2 industrial partners willing to run operational pilots at cost-sharing. Target: (1) a power utility company in Tamil Nadu/Karnataka with >500 km of transmission lines requiring quarterly inspection; (2) a pipeline operator in

Rajasthan/Gujarat. Provide the complete MVP fleet (2W + 1C + 1K1) at near-cost pricing (~$8,000 per deployment) in exchange for: unrestricted operational data access, a case study publication right, a letter of intent for production fleet purchase, and on-site support access for R&D.;

Commercial value proposition for pilot partner: a drone fleet that covers 50+ km of line in a single 7-hour deployment day vs 3–4 days with single drones. ROI justification: at ■15,000/day for a drone operator team, the fleet pays back its ■6.5L hardware cost in approximately 43 inspection days (assuming 30% faster coverage vs single-drone approach).

7.2 Phase 2 — Product Launch and SaaS (Month 12–24)

- **Fleet OS SaaS launch:** GCS software with CBBA-SOC, mission planner, analytics dashboard. Pricing: ■15,000/drone/year base subscription. Target 10 fleets × 5 drones = ■75L ARR.
- **Mission package marketplace:** Agriculture survey pack, inspection pack, SAR pack. One-time licence ■50,000–■2,00,000 per package per fleet.
- **Hardware production:** Deliver first 50 production units at ■80,000/worker ($960 at current rates).
- **Dealer channel:** Sign 3 drone-service-provider dealers in Maharashtra, Gujarat, Karnataka for sales and Level 1 support. 15% dealer margin; we provide training and spare parts programme.
- **DGCA compliance:** File for BVLOS experimental permit (Form D-3) during Phase 1 operations. Targeted approval for BVLOS operations in rural areas by Month 18.

7.3 Phase 3 — Scale and International (Month 24–48)

- Apply for Series A funding at ■15–20 Cr (≈$1.8–2.4M) on ■3Cr+ ARR.
- Enter Singapore market (MAS/CAAS drone framework is permissive; English-speaking; gateway to ASEAN).
- File PCT patent applications in US, EP, IN, SG, AU.
- OEM partnership: approach one large agricultural drone company (IFFCO, Dhanuka group) for white-label supply.
- Defence: submit technology demonstration proposal to DRDO under iDEX (Innovations for Defence Excellence) scheme.

8. FINANCIAL PROJECTIONS

Year	Fleets Deployed	Hardware Rev	SaaS ARR	IP/Other	COGS	Gross Profit	GP Margin
Y1 (Pilot)	2 pilot fleets	■16L	■0	■0	■14L	■2L	12%
Y2 (Launch)	10 fleets (50 drones)	■1.2Cr	■75L	■10L	■1.1Cr	■75L	36%
Y3 (Growth)	35 fleets (175 drones)	■3.5Cr	■2.6Cr	■40L	■3.1Cr	■3.4Cr	50%
Y4 (Scale)	80 fleets (400 drones)	■6.4Cr	■6.0Cr	■80L	■5.8Cr	■7.4Cr	56%
Y5 (Platform)	150 fleets (750 drones)	■9Cr	■11.25Cr	■1.5Cr	■8Cr	■13.75Cr	63%

Table 8. Five-year financial projections (INR). All revenue in Indian Rupees crore (Cr) / lakh (L). 1 Cr = ■1,00,00,000.

9. REGULATORY COMPLIANCE PATHWAY

Requirement	Authority	Status	Timeline	Cost Est.
RPAS registration (each drone)	DGCA Digital Sky Portal	Required immediately for any flight	2–4 weeks per drone	■1,000/drone
UIN (Unique Identification)	DGCA	Required for each drone	Immediate (online)	■500/drone
Remote Pilot Certificate (RPC)	DGCA approved training org.	Required for all operators	4–6 weeks training	■25,000–35,000/operator
Manufacturer type certificate	DGCA	Required for commercial sale	6–12 months application	■2L–5L
BVLOS experimental permit (Form D-3)	DGCA + AAI	For Phase 1 pilot operations	File at Month 3; approve Month 6–9	■50,000 filing
BVLOS commercial approval	DGCA	For Phase 2 commercial operations	File at Month 12; approve Month 18–24	■1L–3L + insurance
FCC Part 15 (US market)	FCC USA	Required for US sales	8–12 weeks	$500–2,000
CE mark (EU market)	CE/Notified Body EU	Required for EU sales	12–16 weeks	€3,000–8,000
Liability insurance	IRDAI-approved insurer	Mandatory for commercial ops	Ongoing annual	■1L–3L/yr per fleet

Table 9. Regulatory compliance requirements and pathway.